

TRY IT YOURSELF

10-4. Guest: Write a program that prompts the user for their name. When they respond, write their name to a file called *guest.txt*.

10-5. Guest Book: Write a while loop that prompts users for their name. Collect all the names that are entered, and then write these names to a file called *guest_book.txt*. Make sure each entry appears on a new line in the file.

Exceptions

Python uses special objects called *exceptions* to manage errors that arise during a program's execution. Whenever an error occurs that makes Python unsure of what to do next, it creates an exception object. If you write code that handles the exception, the program will continue running. If you don't handle the exception, the program will halt and show a *traceback*, which includes a report of the exception that was raised.

Exceptions are handled with try-except blocks. A *try-except* block asks Python to do something, but it also tells Python what to do if an exception is raised. When you use try-except blocks, your programs will continue running even if things start to go wrong. Instead of tracebacks, which can be confusing for users to read, users will see friendly error messages that you've written.

Handling the ZeroDivisionError Exception

Let's look at a simple error that causes Python to raise an exception. You probably know that it's impossible to divide a number by zero, but let's ask Python to do it anyway:

*division
_calculator.py*

```
print(5/0)
```

Python can't do this, so we get a traceback:

```
Traceback (most recent call last):  
  File "division_calculator.py", line 1, in <module>  
    print(5/0)  
    ~~~
```

❶ ZeroDivisionError: division by zero

The error reported in the traceback, `ZeroDivisionError`, is an exception object ❶. Python creates this kind of object in response to a situation where it can't do what we ask it to. When this happens, Python stops the program and tells us the kind of exception that was raised. We can use this information to modify our program. We'll tell Python what to do when this kind of exception occurs; that way, if it happens again, we'll be prepared.